

A convenient bioinformatic tools with proof of concept results regarding the repertoire profile of mice and human patients allergic to anesthetic agents



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Summary

INTRODUCTION. The immune system generates a vast repertoire of antibodies in response to infection or immunization. Exploration of such repertoire is made possible by sorting antigen-specific memory B cells by FACS, or, as recently published by our group, by sorting antigen-specific antibody-secreting plasmablasts and plasma cells by droplet-based microfluidic sorting (drop-seq). Such technologies allow to pair antigen specificity with antibody gene sequences (VH-JH and VL-JL) but imply to tackle ever growing datasets for the bioinformatic analysis of antibody repertoires.

OBJECTIVE: Develop several bioinformatic tools for analyzing the output of such repertoires, freely accessible to the scientific community and upgraded to integrate the complexity of VH-VL pairing in the analyses, tree generation and antibody maturation analyses, in agreement with the Adaptive Immune Receptor Repertoire (AIRR) standards and recommendations.

METHOD. The one currently in development is Repertoire Profiler (https://github.com/gael-millot/repertoire_profiler). It benefits from the R Immcantation solution (<https://immcantation.readthedocs.io>) and performs:

- (1) annotation of mRNA sequencing of the immunoglobulin heavy (VH) or light variable (VL) region,
- (2) clustering of the annotated sequences into clonal groups,
- (3) visualization of the clonal groups in alignments and in trees integrating potential metadata, like affinities for the antigen,
- (4) statistical description of the repertoire.

Output files include the matrix count of the V and J allele usage (repertoire), that can be analyzed with our Comat tool (<https://gitlab.pasteur.fr/gmillot/comat>). Comat performs the 2 by 2 statistical comparison of repertoires of same dimension in batch and regroup the results of the batch into a multidimensional scale analysis.

RESULTS. As a proof-of-concept we analyzed here plasma cell repertoires from mice and memory B cell repertoires from patients, using as an antigen rocuronium, a neuromuscular blocking agent used during surgery and responsible for allergic reactions. Rocuronium-specific VH sequences from mouse plasma cells demonstrated an oligoclonal repertoire with the same predominant V(D)J rearrangement in two different mice. All antibodies we expressed were of high affinity for rocuronium. Rocuronium-specific VH sequences from human memory B cells demonstrated a highly polyclonal repertoire with no detectable clonal group nor common V(D)J rearrangement between patients. All antibodies we expressed were of low affinity for rocuronium.

CONCLUSIONS. Repertoire Profiler successfully analyzed and represented VH repertoires from mouse plasma cells and human memory B cells.

Metadata could be associated to VH analyses in clonal groups.

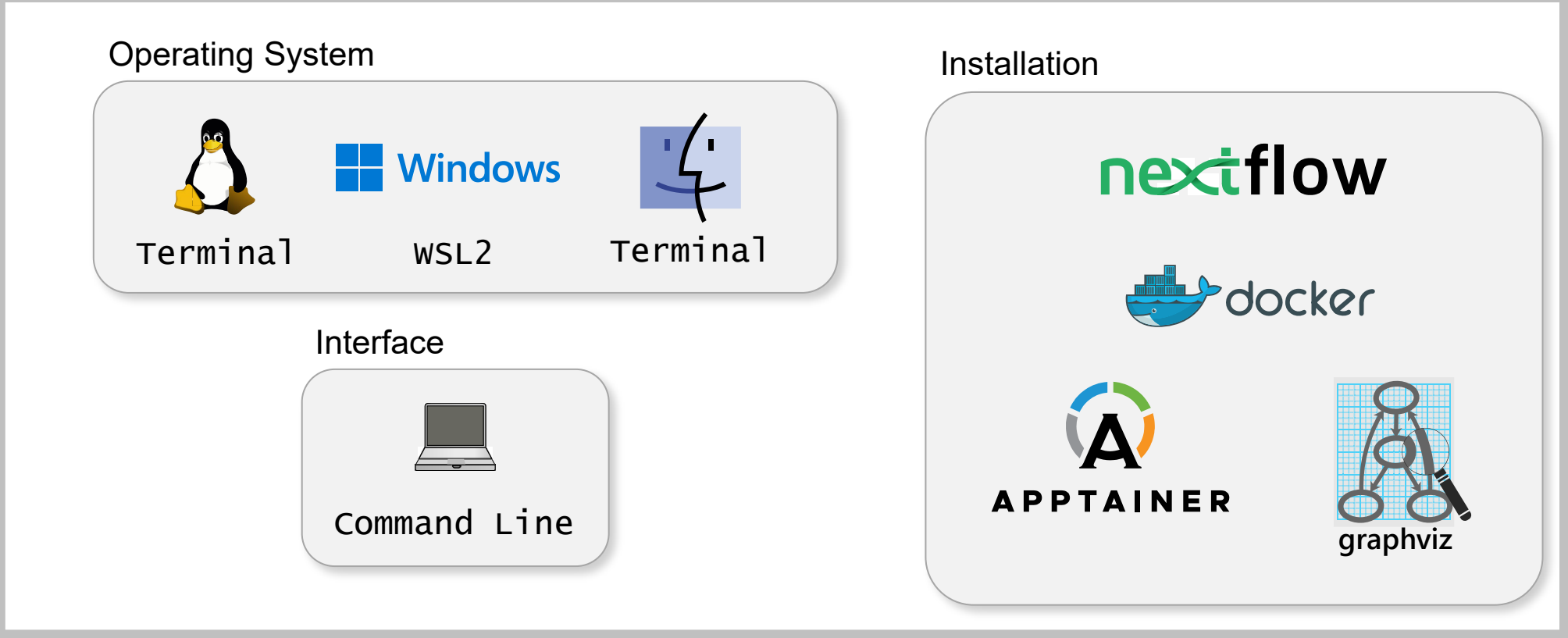
The oligoclonal versus polyclonal anti-rocuronium repertoires may be related to the mode of immunization/sensitization.

PERSPECTIVES. Develop combined VH-VL repertoires for in-depth analyses of clonal groups.

Biological question

Intra and inter individual diversity of antigen-specific repertoire ?
Relation between VJ alleles usage and antigen affinity ?

Requirement



Execution

1) Download files at https://github.com/gael-millot/repertoire_profiler.

2) Put the files into a specific folder.



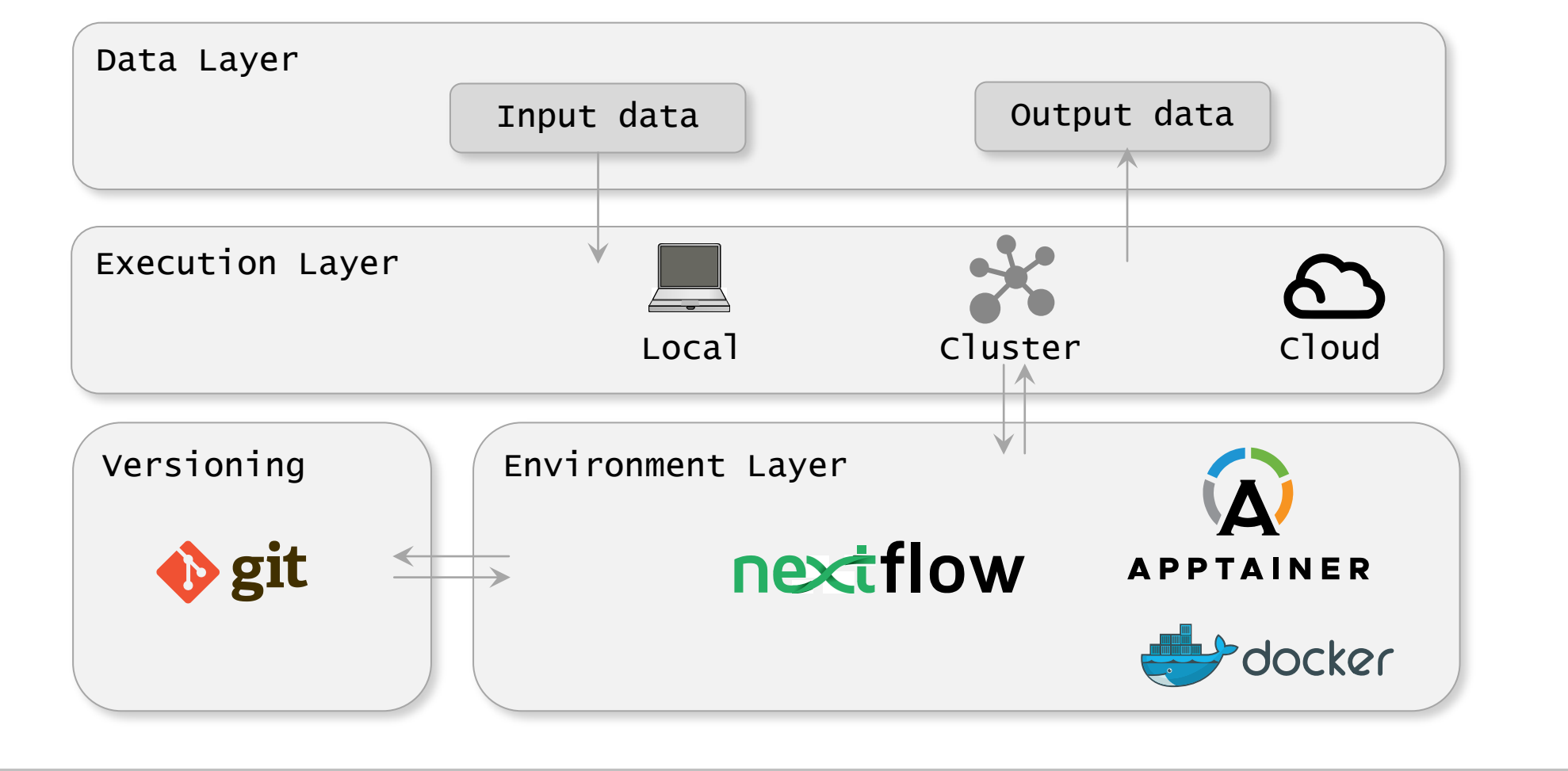
3) Open the repertoire_profiler.config file and set the parameters:

```
env {
  sample_path = "https://zenodo.org/records/14580292/files/ig_clustering_test_1_VH.zip" // single character string of the path of the fasta files directory. The last / can be added or not, as it is
  // removed by nextflow file(). Can also be a .zip file that contains only fasta files. Warning: the fasta names must not start by a digit, before alakazam:readChangeOb() function is fixed by the
  // maintainer. Example : sample_path="/mnt/c/Users/gmillot/Documents/Git_projects/repertoire_profiler/dataset/ig_clustering_test_1_VH" or sample_path="/pasteur/apptainer/homes/gmillot/dataset/
  // 20210707_AV07016_HAD-III-89_plate3_IgK_sanger_seq". Example with spaces in the path: sample_path="/mnt/c/ROCURONIUM PROJECT/01 Primary data/04.Repertoire analysis/SORT1/SORT1 Seq-original/
  // xlsx_to_fasta_1669018924/All/VL". Example: sample_path = "/mnt/c/Users/gmillot/Documents/Git_projects/repertoire_profiler/dataset/ig_clustering_test_1_VH". Example: sample_path = "https://zenodo.
  // org/records/14580292/files/ig_clustering_test_1_VH.zip". Example with unaligned seq by Igblast and unproductive seq: sample_path = "/mnt/c/Users/gmillot/Documents/Git_projects/repertoire_profiler/
  // dev/dataset/ip_cambodia unaligned unproductive seq".
  meta_path = "https://zenodo.org/records/14771586/files/metadata.tsv" // single character string of a valid path of a metadata file for adding info to the results. Write "NULL" if no metadata to add.
  // WARNING: the metadata .tsv table must include a first column named "Label" containing sequence names, i.e., the header of some of the fasta files from sample_path, without the ">" of the header.
  // Additional columns (quant or qual) can then be added after the first column to modify the leafs of the tree. For instance: "KD", or Antibody name. Example: meta_path = "/mnt/c/Users/gmillot/
  // Documents/Git_projects/repertoire_profiler/dataset/metadata.tsv". Example: meta_path = "NULL". Example: meta_path = "https://zenodo.org/records/14580245/files/metadata.tsv".
  meta_seq_names = "Seq_ID" // single character string of the column name of the file indicated in the meta_path parameter. This column name must contain the header of some of the fasta files from
  // sample_path, without the ">" of the header. Cannot be "NULL". Ignored if meta_path is "NULL". Example: meta_seq_names = "Label".
}
```

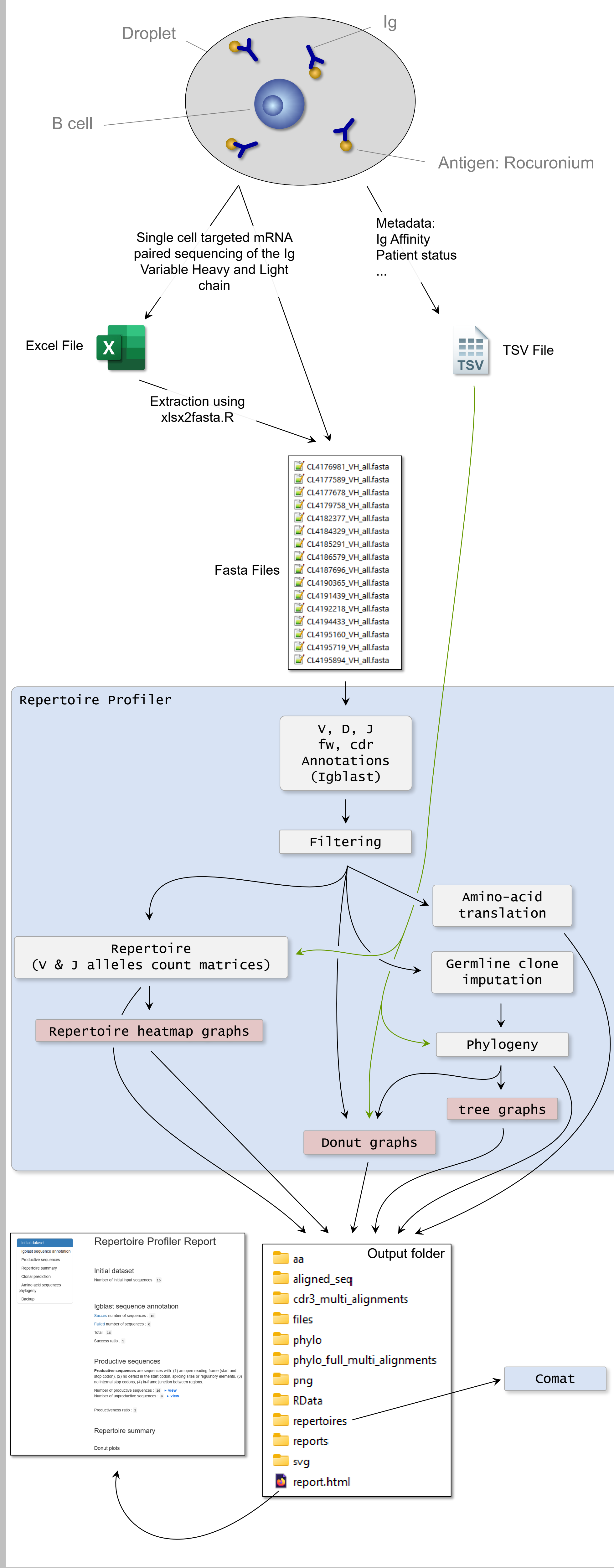
4) Execute this command line: `nextflow run -c nextflow.config main.nf`

Quality

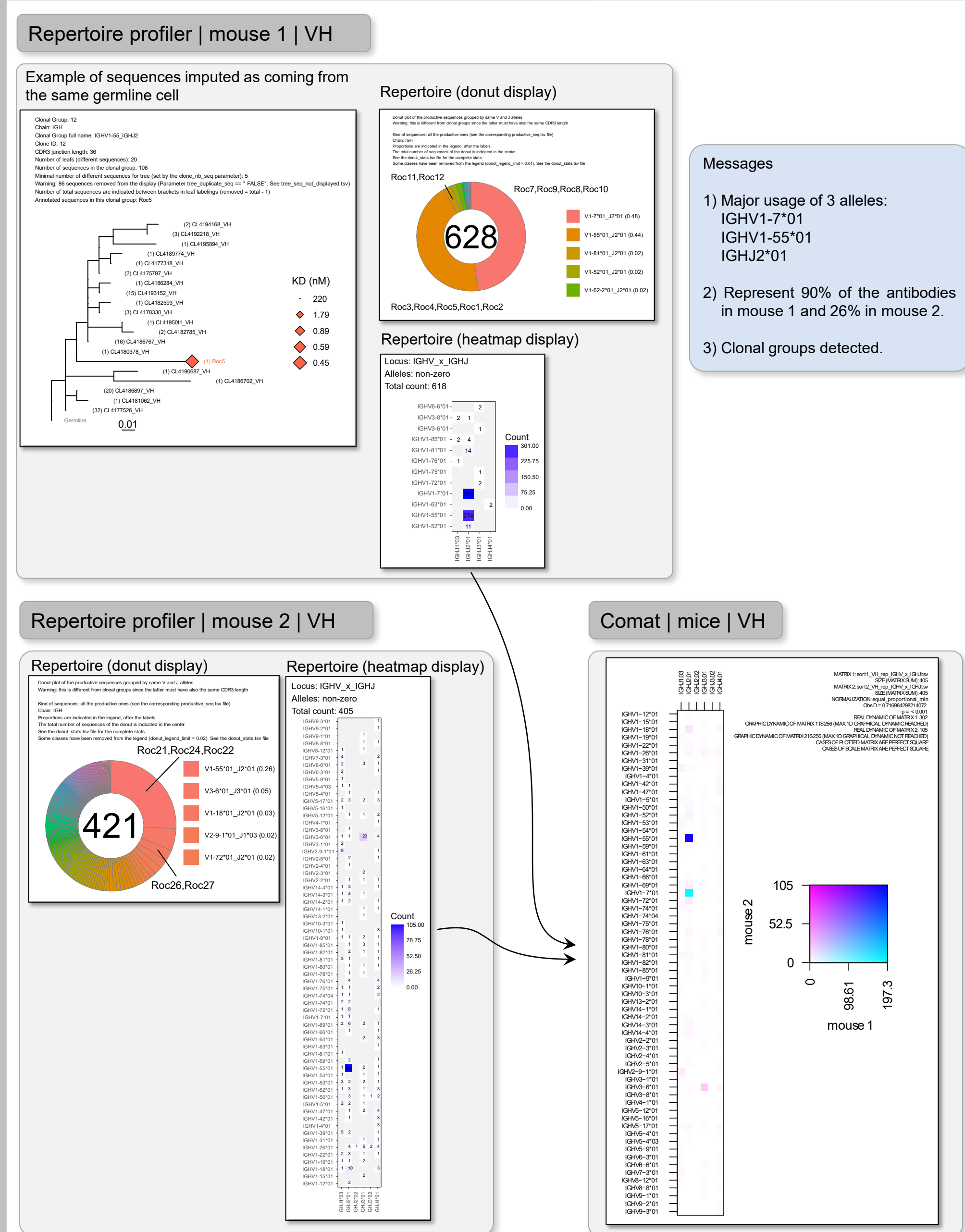
FAIR (Findable, Accessible, Interoperable, Reusable).
Reproducible.



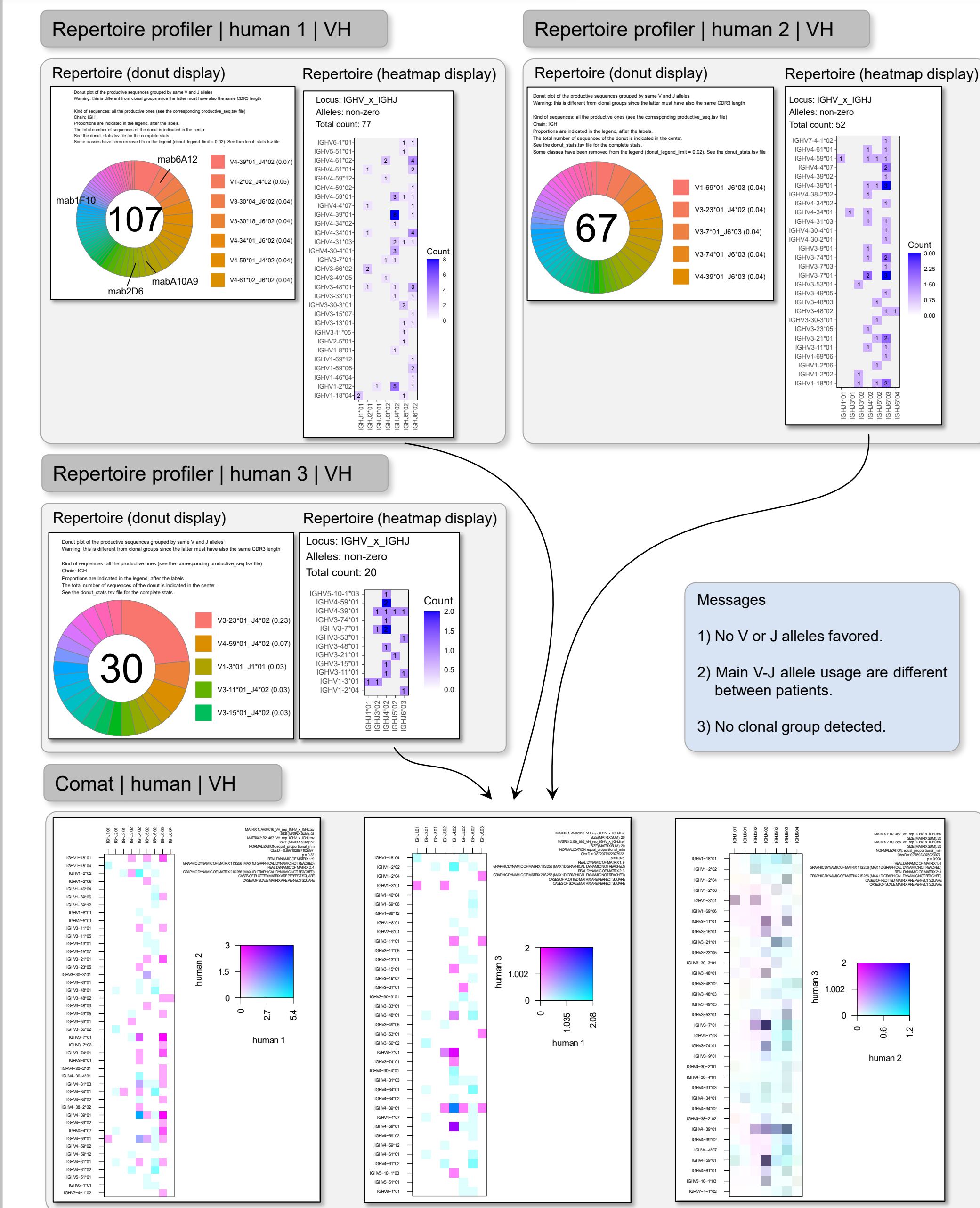
Pipeline



Results: mouse



Results: human



Publications

Antibody-secreting cell repertoires hold high-affinity anti-rocuronium specificities that can induce anaphylaxis in vivo. Dejoux A, Zhu Q, Woolfe A, Godon O, Ellouze S, Mottet G, Castillon C, Gills C, Pecavel C, Ganneau C, Iannascoli B, Lemoine F, Sau F, England P, Reber LL, Goulet-Chéron A, de Chaisemartin L, Haouz A, Millot GA, Bay S, Gérard A, Jonsson F, Chollet-Martin S, Bruhns P. J Allergy Clin Immunol. 2023; Jan 30:S0091-6748(23)01157-7. doi: 10.1016/j.jaci.2023.01.025. Online ahead of print. PMID: 36800258

Rocuronium-specific antibodies drive perioperative anaphylaxis but can also function as reversal agents in preclinical models. Dejoux A, Zhu Q, Ganneau C, Goff OR, Godon O, Lemaître J, Relouzat F, Huetz F, Sokal A, Vandenberghe A, Pecavel C, Hanaoui L, Derenne J, Gills CM, Iannascoli B, Wang Y, Rose T, Mertens C, Nicaise-Roland P, NASA Study Group, England P, Mahévas M, de Chaisemartin L, Le Grand R, Lüscher H, Saul F, Pissis C, Haouz A, Reber LL, Chappert P, Jonsson F, Ebo DG, Millot GA, Bay S, Chollet-Martin S, Goulet-Chéron A, Bruhns P. Sci Transl Med. 2024 Sep 11;16(76):eado4463. doi: 10.1126/scitranslmed.ado4463